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- Hierarchical models
 - Rats example
 - Factory example
- Exchangeability
- Computational aspects
- Additional material
 - 8 schools example

Bayesian Statistics and Data Analysis

Lecture 7

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Thanks to Aki Vehtari, Aalto University



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Section 1

Hierarchical models



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Stan tutorial for hierarchical models

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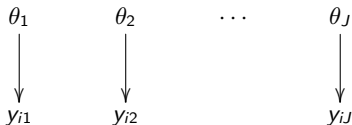
- Before Assignment 7, watch [Hierarchical Modelling in Stan: Predicting the Premier League](#)
- While watching, connect the example to:
 - partial pooling
 - group-level parameters
 - comparison with non-hierarchical models
 - Stan diagnostics for hierarchical models



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Survival probabilities across hospitals

- Example: comparing hospitals
 - Hospital j : survival probability θ_j
 - Patient i 's survival outcome: y_{ij}

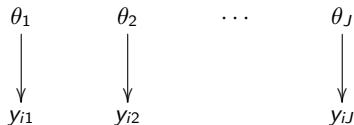




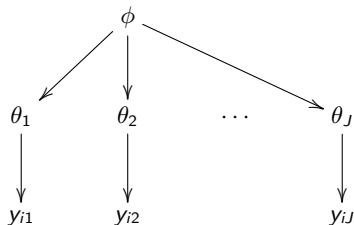
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Survival probabilities across hospitals

- Example: comparing hospitals
 - Hospital j : survival probability θ_j
 - Patient i 's survival outcome: y_{ij}



- Model the θ_j as similar, not identical



- A shared population distribution has hyperparameters ϕ
- Both θ_j and ϕ are unknown



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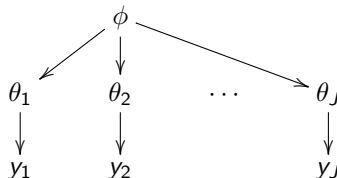
- $j = 1, \dots, J$: groups; $i = 1, \dots, n_j$: observations in group j
- y_{ij} : individual observation; y_j : group data or a group summary; $y = (y_1, \dots, y_J)$
- θ_j : group-specific parameter; $\theta = (\theta_1, \dots, \theta_J)$
- ϕ : population hyperparameters

For normal hierarchies:

- μ : population mean; τ : between-group SD
- σ : within-group SD of individual observations



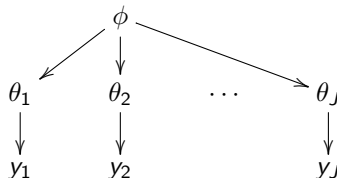
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- Given ϕ , the θ_j are independent with a common population distribution $p(\theta_j | \phi)$



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- Given ϕ , the θ_j are independent with a common population distribution $p(\theta_j | \phi)$
- Given θ , groups are independent, and y_j depends only on its own parameter θ_j
- The hyperprior $p(\phi)$ describes uncertainty about the population distribution



Bayes' rule and the conditional-independence assumptions give

$$\begin{aligned} p(\theta, \phi \mid y) &\propto p(y \mid \theta) p(\theta \mid \phi) p(\phi) \\ &\propto p(\phi) \prod_{j=1}^J [p(\theta_j \mid \phi) p(y_j \mid \theta_j)] . \end{aligned}$$

- One hyperprior; each group contributes a population factor and a likelihood

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Bayes' rule and the conditional-independence assumptions give

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- One hyperprior; each group contributes a population factor and a likelihood
- Given ϕ and y , the group parameters are independent
- Averaging over the shared, uncertain ϕ generally makes the θ_j dependent in the posterior
- Each group's data inform ϕ , so other groups can inform its estimate of θ_j

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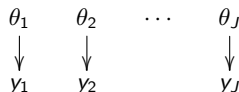
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No, complete, and partial pooling

No pooling (separate)
Independent priors; each θ_j uses
only group j 's data.





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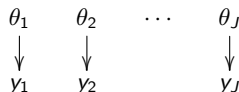
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No, complete, and partial pooling

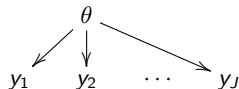
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Independent priors; each θ_j uses only group j 's data.



Complete pooling (pooled)

All groups inform one common θ .





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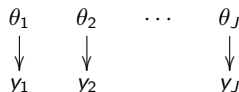
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No, complete, and partial pooling

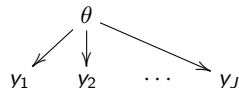
No pooling (separate)

Independent priors; each θ_j uses only group j 's data.



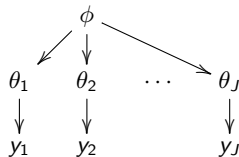
Complete pooling (pooled)

All groups inform one common θ .



Partial pooling (hierarchical)

Distinct θ_j share information through ϕ .





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Pair discussion: group size and pooling

Discuss in pairs (1–2 minutes).

Two groups have the same observed success rate:

$$\text{Group A: } \frac{1}{5} = 20\%, \quad \text{Group B: } \frac{100}{500} = 20\%.$$

Both share the same population information, with a population mean of 32%.

Which estimate should move more toward the population mean? Why?



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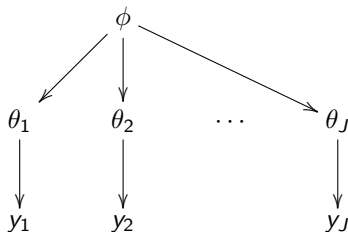
Answer: Group A. Its five observations provide weaker evidence, so the population information receives more weight.



Predictive distributions for hierarchical models

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- Two types of predictive distributions
 1. A new observation in **an existing group**
 2. A new observation in **a new group**





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Hierarchical binomial model: rats

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- Medicine testing
- Type F344 female rats in control group given placebo
 - count how many get endometrial stromal polyps (benign (noncancerous) growths in the uterus)
 - familiar binomial model example



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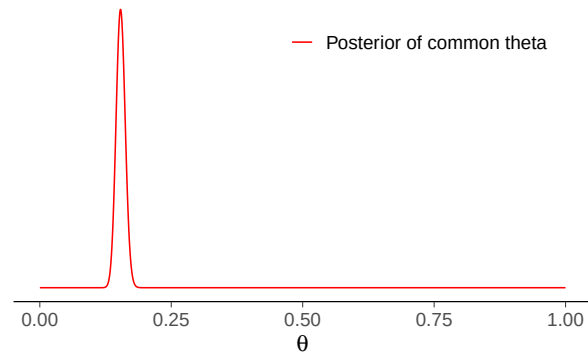
- Medicine testing
- Type F344 female rats in control group given placebo
 - count how many get endometrial stromal polyps (benign (noncancerous) growths in the uterus)
 - familiar binomial model example
- Experiment has been repeated 71 times

0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/19	0/19	0/19
0/19	0/18	0/18	0/17	1/20	1/20	1/20	1/20	1/19	1/19
1/18	1/18	2/25	2/24	2/23	2/20	2/20	2/20	2/20	2/20
2/20	1/10	5/49	2/19	5/46	3/27	2/17	7/49	7/47	3/20
3/20	2/13	9/48	10/50	4/20	4/20	4/20	4/20	4/20	4/20
4/20	10/48	4/19	4/19	4/19	5/22	11/46	12/49	5/20	5/20
6/23	5/19	6/22	6/20	6/20	6/20	16/52	15/46	15/47	9/24
4/14									



Hierarchical binomial model: rats

Pooled model

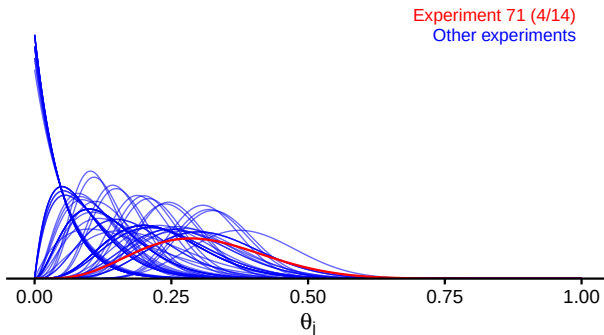


Complete pooling assumes one common probability for all experiments.



Separate model

Existing-group posteriors

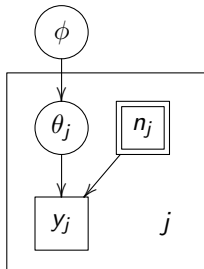


No pooling estimates each probability separately, without learning a shared population distribution.



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Reading the model diagrams



- Circle: unknown quantity
- Single box: observed quantity or a supplied estimate
- Double box: fixed, given quantity
- Arrow: the child's distribution depends on the parent
- Outer box (**plate**): repeat its contents for $j = 1, \dots, J$

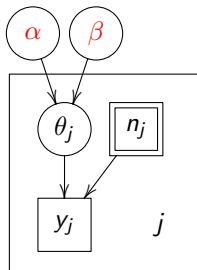
These are the conventions used in this lecture. A nested i plate repeats observations within each group.



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Hierarchical binomial model: rats

- Here $\phi = (\alpha, \beta)$; y_j counts affected rats out of known n_j



$$\theta_j \mid \alpha, \beta \sim \text{Beta}(\alpha, \beta)$$

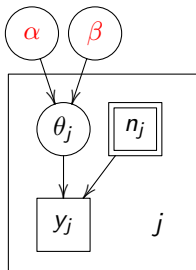
$$y_j \mid \theta_j, n_j \sim \text{Binomial}(n_j, \theta_j)$$



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Hierarchical binomial model: rats

- Here $\phi = (\alpha, \beta)$; y_j counts affected rats out of known n_j



$$\theta_j \mid \alpha, \beta \sim \text{Beta}(\alpha, \beta)$$

$$y_j \mid \theta_j, n_j \sim \text{Binomial}(n_j, \theta_j)$$

$$p(\theta, \alpha, \beta \mid y) = \left[\prod_{j=1}^J p(\theta_j \mid \alpha, \beta, y_j) \right] p(\alpha, \beta \mid y)$$

- One hyperposterior, not one per group
- The θ_j are independent given α, β, y , but generally dependent after averaging over α, β



For fixed $\alpha, \beta > 0$, the binomial likelihood gives

$$\begin{aligned} p(\theta_j \mid \alpha, \beta, y_j) &\propto \theta_j^{y_j} (1 - \theta_j)^{n_j - y_j} \theta_j^{\alpha - 1} (1 - \theta_j)^{\beta - 1} \\ &\propto \theta_j^{\alpha + y_j - 1} (1 - \theta_j)^{\beta + n_j - y_j - 1}. \end{aligned}$$

Thus

$$\theta_j \mid \alpha, \beta, y_j \sim \text{Beta}(\alpha + y_j, \beta + n_j - y_j).$$

- The population distribution supplies α, β
- Group j supplies y_j successes and $n_j - y_j$ failures

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Beta-binomial shrinkage

Write the population mean and concentration as

$$m = \frac{\alpha}{\alpha + \beta}, \quad \kappa = \alpha + \beta.$$

For $n_j > 0$,

$$\begin{aligned} E(\theta_j \mid \alpha, \beta, y_j) &= \frac{\alpha + y_j}{\alpha + \beta + n_j} \\ &= \underbrace{\frac{n_j}{n_j + \kappa}}_{\text{data weight}} \frac{y_j}{n_j} + \underbrace{\frac{\kappa}{n_j + \kappa}}_{\text{population weight}} m. \end{aligned}$$

- The conditional posterior mean lies between y_j/n_j and m
- With α, β fixed, smaller groups receive more population weight
- Full Bayes also averages over uncertainty in α, β



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- Population prior $\text{Beta}(\theta_j | \alpha, \beta)$
- Hyperprior $p(\alpha, \beta)$?
 - α, β both affect the location and scale
 - BDA3 (pp. 110–111) uses, for $\alpha, \beta > 0$,
 $p(\alpha, \beta) \propto (\alpha + \beta)^{-5/2}$
 - This hyperprior is **improper**: its integral is infinite
 - BDA shows that the posterior is proper for these rat data; posterior propriety must be checked when using an improper prior
 - An example-specific choice, not a general default



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Hierarchical inference with posterior draws

For $s = 1, \dots, S$:

1. Draw $(\alpha^{(s)}, \beta^{(s)}) \sim p(\alpha, \beta \mid y)$
2. For every group j , draw independently conditional on this pair:

$$\theta_j^{(s)} \sim \text{Beta}(\alpha^{(s)} + y_j, \beta^{(s)} + n_j - y_j)$$

3. In Stan we usually get θ_j right away as a parameter

Share one hyperparameter draw across all groups at each step.

$$\mathbb{E}[h(\theta_j) \mid y] = \int h(\theta_j) p(\theta_j \mid y) d\theta_j \approx \frac{1}{S} \sum_{s=1}^S h(\theta_j^{(s)}).$$

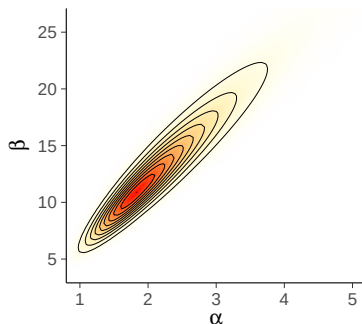
Choosing $h(\theta) = \theta$ estimates a mean; $h(\theta) = \mathbf{1}(\theta > c)$ estimates a probability.



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Rats: learning the population distribution

The marginal of α and β

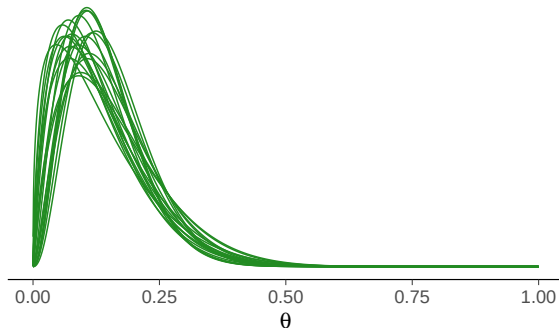


The hyperposterior $p(\alpha, \beta \mid y)$ uses all 71 experiments.



Rats: learning the population distribution

Beta(α, β) given posterior draws of α and β

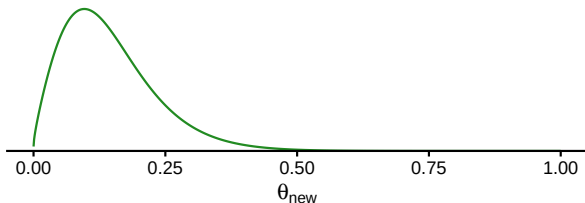


Each curve is Beta(α, β) for one posterior draw of (α, β) .



New group's probability

Posterior predictive $p(\theta_{\text{new}} | y)$



$$p(\theta_{\text{new}} | y) = \int_0^\infty \int_0^\infty p(\theta_{\text{new}} | \alpha, \beta) p(\alpha, \beta | y) d\alpha d\beta.$$

This averages the population curves over the hyperposterior. It predicts a new experiment's probability, before observing its data.

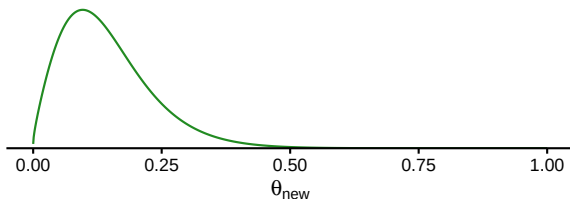


Pair discussion: a new rat experiment

Discuss in pairs (1–2 minutes).

New group's probability

Posterior predictive $p(\theta_{\text{new}} | y)$



A new experiment will include n_{new} rats.

How would you use this distribution to predict the number of affected rats, y_{new} ? Describe the simulation steps.

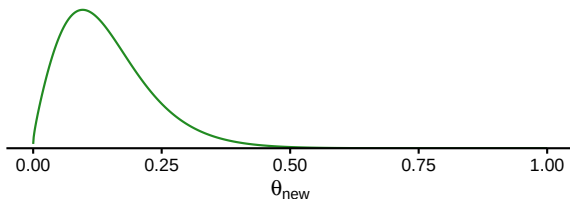


Pair discussion: a new rat experiment

Discuss in pairs (1–2 minutes).

New group's probability

Posterior predictive $p(\theta_{\text{new}} | y)$



Answer: Draw a probability from the green distribution, then draw a count:

$$\theta_{\text{new}}^{(s)} \sim p(\theta_{\text{new}} | y), \quad y_{\text{new}}^{(s)} \sim \text{Binomial}(n_{\text{new}}, \theta_{\text{new}}^{(s)}).$$

Repeat and summarize the simulated counts.



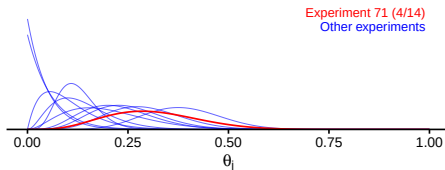
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Rats: partial pooling

Experiment 71 (4/14), shown in red.

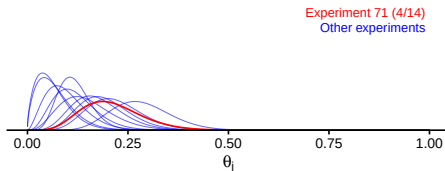
Separate model

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Hierarchical model

Existing-group posteriors



Experiment 71 moves toward the population.
Both plots use the same experiments and scales.

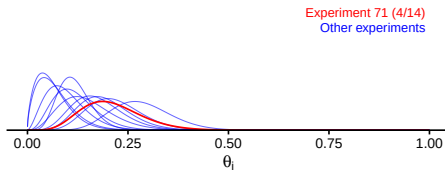


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Rats: existing and new experiments

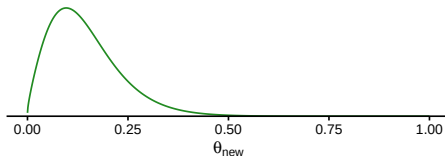
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Existing-group posteriors



New group's probability

Posterior predictive $p(\theta_{\text{new}} | y)$



Existing $p(\theta_j | y)$ uses group j 's data as well as the population.
New $p(\theta_{\text{new}} | y)$ has no new-group data.



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Pair discussion: predictions for rats

Discuss in pairs (1–2 minutes).

What predictive distributions can we have?

Think about both an existing experiment and a new similar experiment.

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Discuss in pairs (1–2 minutes).

What predictive distributions can we have?

Think about both an existing experiment and a new similar experiment.

- Future outcomes or counts in an existing experiment, using $p(\theta_j | y)$
- A new experiment's probability, $p(\theta_{\text{new}} | y)$
- Outcomes or counts in that new experiment, also including binomial sampling variation



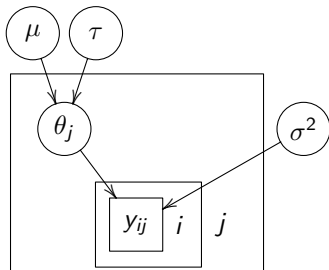
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Hierarchical normal model: factory

- We evaluate the quality produced by six machines in a factory
- Assume hierarchical model
 - each machine has its own (average) quality θ_j and **common variance** σ^2

$$\theta_j | \mu, \tau \sim \mathcal{N}(\mu, \tau^2)$$

$$y_{ij} | \theta_j \sim \mathcal{N}(\theta_j, \sigma^2)$$



- Can be used to predict the future quality produced by each machine and quality produced by a new similar machine



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Factory model: a complete hierarchy

One illustrative choice of **proper, independent priors** is

$$\begin{aligned}\mu &\sim \mathcal{N}(m_0, s_0^2), \\ \tau &\sim \text{HalfNormal}(0, A_\tau), \quad \sigma \sim \text{HalfNormal}(0, A_\sigma).\end{aligned}$$

The common-variance factory model then has

$$\begin{aligned}\theta_j \mid \mu, \tau &\stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \tau^2), & j = 1, \dots, J, \\ y_{ij} \mid \theta_j, \sigma &\sim \mathcal{N}(\theta_j, \sigma^2), & i = 1, \dots, n_j.\end{aligned}$$

- $\text{HalfNormal}(0, A)$: a $\mathcal{N}(0, A^2)$ restricted to $\mathbb{R}^+ = (0, \infty)$
- Observations are independent given the group means and σ
- Fix $m_0, s_0, A_\tau, A_\sigma$ using subject knowledge (e.g. prior predictive distribution)



A half-normal prior for a standard deviation

A simple way to restrict τ to positive values is

$$Z \sim \mathcal{N}(0, A_\tau^2), \quad \tau = |Z|.$$

This defines $\tau \sim \text{HalfNormal}(0, A_\tau)$ on $\mathbb{R}^+ = (0, \infty)$, with

$$\Pr(\tau < 1.96A_\tau) \approx 0.95.$$

- The 0 denotes the mean of the underlying normal Z , not of τ
- A_τ has the same units as the outcome
- Example: to put 95% prior probability below a between-machine SD of 20 quality units, choose $A_\tau \approx 20/1.96 = 10.2$
- Choose A_σ separately for variation within a machine
- These scales are illustrative, not universal defaults



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The prior on variation affects pooling

- Small τ : machine means lie close to μ , encouraging stronger pooling
- Large τ : machine means can differ more, allowing less pooling
- With few groups, the data may give little information about τ , so its prior can matter substantially
- Check prior predictive outcomes and compare plausible prior scales
- A uniform prior on τ , on τ^2 , or on $\log \tau$ gives different assumptions; “flat” does not mean neutral

BDA3, Section 5.7 discusses priors for hierarchical variation. The half-normal specification here is one modelling choice.



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Factory: observations and means

- In the common-variance model, observations are independent given θ_j and σ :

$$y_{ij} \mid \theta_j, \sigma \sim \mathcal{N}(\theta_j, \sigma^2), \quad i = 1, \dots, n_j.$$

- Machine j 's sample mean and its sampling variance are

$$\bar{y}_{.j} = \frac{1}{n_j} \sum_{i=1}^{n_j} y_{ij},$$

$$\text{Var}(\bar{y}_{.j} \mid \theta_j, \sigma) = \frac{\sigma^2}{n_j} = s_j^2.$$

SD of observations: σ ; SE of the mean: $s_j = \sigma / \sqrt{n_j}$.



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Factory: observations and means

- In the common-variance model, observations are independent given θ_j and σ :

$$y_{ij} \mid \theta_j, \sigma \sim \mathcal{N}(\theta_j, \sigma^2), \quad i = 1, \dots, n_j.$$

- Machine j 's sample mean and its sampling variance are

$$\bar{y}_{.j} = \frac{1}{n_j} \sum_{i=1}^{n_j} y_{ij},$$

$$\text{Var}(\bar{y}_{.j} \mid \theta_j, \sigma) = \frac{\sigma^2}{n_j} = s_j^2.$$

SD of observations: σ ; SE of the mean: $s_j = \sigma / \sqrt{n_j}$.

- Thus $\bar{y}_{.j} \mid \theta_j, \sigma \sim \mathcal{N}(\theta_j, s_j^2)$. We condition on σ here; full Bayes also learns σ from the data.



Normal shrinkage: the weights

For fixed μ , $\tau > 0$ and $\sigma > 0$, let $s_j = \sigma / \sqrt{n_j}$.

$$E(\theta_j \mid \mu, \tau, \sigma, y) = w_j \bar{y}_{.j} + (1 - w_j) \mu, \quad w_j = \frac{\tau^2}{\tau^2 + s_j^2}.$$

- A precise machine mean (small s_j) receives more weight
- Small between-machine variation τ gives more weight to μ
- As $\tau \rightarrow 0$, the conditional mean approaches μ ; as $\tau \rightarrow \infty$, it approaches $\bar{y}_{.j}$
- In full Bayes, average over $p(\mu, \tau, \sigma \mid y)$
- Strong pooling makes the machine means similar; their common value can still be uncertain because μ is unknown

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Normal shrinkage: a numerical example

Suppose machine 1 has $\bar{y}_{\cdot 1} = 28$ from $n_1 = 4$ observations. For illustration, **fix** $\mu = 8$, $\tau = 5$ and $\sigma = 30$, so $s_1 = 30/\sqrt{4} = 15$.

$$w_1 = \frac{5^2}{5^2 + 15^2} = \frac{25}{250} = 0.10,$$

$$m_1 = 0.10 \cdot 28 + 0.90 \cdot 8 = 10,$$

$$v_1 = \left(\frac{1}{225} + \frac{1}{25} \right)^{-1} = 22.5.$$

The conditional posterior is $\theta_1 \mid \mu, \tau, \sigma, y \sim \mathcal{N}(m_1, v_1)$: mean 10, SD $\sqrt{22.5} \approx 4.74$.

- The noisy machine mean receives 10% of the weight
- The full factory posterior also averages over μ, τ, σ



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Pair discussion: precision and shrinkage

Discuss in pairs (1–2 minutes).

Two machines have the same sample mean, different from μ . They share a within-machine SD σ , but different sample sizes give different standard errors of their means:

$$s_1 = 5, \quad s_2 = 20.$$

Hold μ , $\tau > 0$ and $\sigma > 0$ fixed.

Which machine's estimate is shrunk more toward μ ?
Why?



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$$s_1 = 5, \quad s_2 = 20.$$

Hold μ , $\tau > 0$ and $\sigma > 0$ fixed.

Which machine's estimate is shrunk more toward μ ? Why?

Answer: Machine 2. Its larger standard error means less precise evidence, so its sample mean receives less weight and μ receives more weight.



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Prediction for an existing machine

In the common-variance model, distinguish the mean θ_j from a future observation $\tilde{y}_j$:

$$p(\tilde{y}_j | y) = \int \int p(\tilde{y}_j | \theta_j, \sigma) p(\theta_j, \sigma | y) d\theta_j d\sigma.$$

For each posterior draw s :

1. Take $(\theta_j^{(s)}, \sigma^{(s)})$ from the **same joint posterior draw**
2. Draw $\tilde{y}_j^{(s)} \sim \mathcal{N}(\theta_j^{(s)}, (\sigma^{(s)})^2)$
 - The $\theta_j^{(s)}$ summarize uncertainty about the machine's mean
 - The $\tilde{y}_j^{(s)}$ also include variation in a new observation



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Prediction for a new similar machine

With no observations yet from the new machine,

$$p(\theta_{\text{new}} | y) = \int \int p(\theta_{\text{new}} | \mu, \tau) p(\mu, \tau | y) d\mu d\tau,$$

$$p(\tilde{y}_{\text{new}} | y) = \int \int p(\tilde{y}_{\text{new}} | \theta_{\text{new}}, \sigma) p(\theta_{\text{new}}, \sigma | y) d\theta_{\text{new}} d\sigma.$$

For each posterior draw s :

1. Take $(\mu^{(s)}, \tau^{(s)}, \sigma^{(s)})$ jointly from $p(\mu, \tau, \sigma | y)$
2. Draw the new mean: $\theta_{\text{new}}^{(s)} \sim \mathcal{N}(\mu^{(s)}, (\tau^{(s)})^2)$
3. For a new observation, also draw $\tilde{y}_{\text{new}}^{(s)} \sim \mathcal{N}(\theta_{\text{new}}^{(s)}, (\sigma^{(s)})^2)$

Predicting a new mean stops at step 2; predicting an observation includes step 3.



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Pair discussion: a new machine

Discuss in pairs (1–2 minutes).

We have data from six machines and want to predict for a new, similar machine 7.

- Its mean quality, θ_7
- The quality of its next item, $\tilde{y}_7$

How do the simulation steps differ? What additional uncertainty enters the second prediction?



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Pair discussion: a new machine

Discuss in pairs (1–2 minutes).

We have data from six machines and want to predict for a new, similar machine 7.

- Its mean quality, θ_7
- The quality of its next item, $\tilde{y}_7$

How do the simulation steps differ? What additional uncertainty enters the second prediction?

Answer: Both start by drawing a new machine mean using posterior draws of the population parameters. Predicting an item also adds within-machine variation:

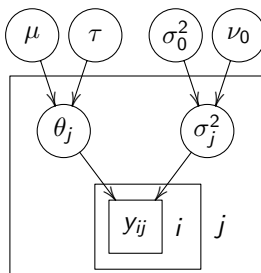
$$\tilde{y}_7^{(s)} \sim \mathcal{N}(\theta_7^{(s)}, (\sigma^{(s)})^2).$$



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Hierarchical normal model: factory

- Six machines with means θ_j and **own variances** σ_j^2



$$y_{ij} \mid \theta_j, \sigma_j^2 \sim \mathcal{N}(\theta_j, \sigma_j^2)$$

$$\theta_j \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2)$$

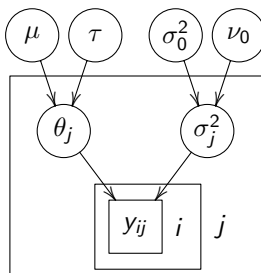
$$\sigma_j^2 \mid \nu_0, \sigma_0^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)$$



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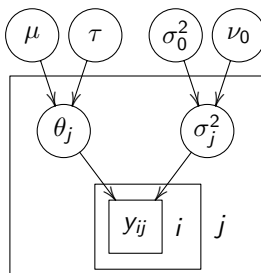
$$\sigma_j^2 \mid \nu_0, \sigma_0^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)$$

- **Pair discussion (1–2 minutes):** What predictive distributions can we have?



Hierarchical normal model: factory

- Six machines with means θ_j and **own variances** σ_j^2



$$y_{ij} \mid \theta_j, \sigma_j^2 \sim \mathcal{N}(\theta_j, \sigma_j^2)$$

$$\theta_j \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2)$$

$$\sigma_j^2 \mid \nu_0, \sigma_0^2 \sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)$$

- **Pair discussion (1–2 minutes):** What predictive distributions can we have?
- Observations from existing or new similar machines



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Section 2

Exchangeability



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Independence and exchangeability

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1. **Independence:** knowing one outcome does not change the distribution of another.
2. **Exchangeability:** swapping the labels or positions of observations leaves their joint distribution unchanged.



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- Motivates a **symmetric joint model**
 - for observations whose labels carry no relevant information
 - for group parameters that we treat symmetrically a priori
- Less restrictive than **IID** (independent and identically distributed)
- IID implies exchangeability, but exchangeability need not imply independence
- Independence alone is not enough: the distributions may differ



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Exchangeability

- *Exchangeability*

Random variables $\theta_1, \dots, \theta_J$ (or $y_1, \dots, y_J$) are **exchangeable** if the joint distribution p is invariant to the permutation of indices

i.e

$$p(\theta_1, \theta_2, \theta_3) = p(\theta_{\pi(1)}, \theta_{\pi(2)}, \theta_{\pi(3)})$$

for any permutation π of the indices. E.g.

$$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$



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$$p(\theta_1, \theta_2, \theta_3) = p(\theta_2, \theta_3, \theta_1)$$

Can we come up with a situation where this doesn't hold?



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Can we come up with a situation where this doesn't hold?
E.g. when we have a trend over indices (like in time series data)



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Can we come up with a situation where this doesn't hold?

E.g. when we have a trend over indices (like in time series data)

- Exchangeability implies symmetry:
If there is no information which can be used *a priori* to separate θ_j from each other, we can assume exchangeability. ("Ignorance implies exchangeability")



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Exchangeability: de Finetti's theorem

- **Bernoulli case:** let $(X_i)_{i=1}^{\infty}$ be an infinite exchangeable sequence with $X_i \in \{0, 1\}$
- There is a random $\theta \in [0, 1]$ such that the X_i are **conditionally IID**:

$$X_i \mid \theta \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$$

- If θ has a density $p(\theta)$, then for every finite J ,

$$p(x_1, \dots, x_J) = \int_0^1 \left[\prod_{i=1}^J \theta^{x_i} (1 - \theta)^{1-x_i} \right] p(\theta) d\theta$$

- The theorem does not choose the mixing distribution: a Beta prior is a modelling choice, not a consequence



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De Finetti's takeaway

Exchangeability provides a justification for modelling observations as independent conditional on a shared unknown parameter, even though they need not be independent marginally.

1. **Usually, we start with a Bayesian model:** an unknown coin bias θ , a prior for θ , and independent flips given θ .
2. **De Finetti starts with symmetry:** the order of the flips carries no information.
3. For an **infinite exchangeable Bernoulli sequence**, the theorem says its distribution must admit exactly that Bayesian representation: a distribution over θ and IID flips conditional on θ .



Coin flips: exchangeable but dependent

Let $\theta \sim \text{Beta}(1, 1)$ and, given θ , let the flips be IID Bernoulli(θ). Write H_i for heads on flip i .

$$\Pr(H_1) = \mathbb{E}(\theta) = \frac{1}{2}.$$

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Coin flips: exchangeable but dependent

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$$\Pr(H_1) = \mathbb{E}(\theta) = \frac{1}{2}.$$

After observing H_1 ,

$$\theta \mid H_1 \sim \text{Beta}(2, 1), \quad \Pr(H_2 \mid H_1) = \mathbb{E}(\theta \mid H_1) = \frac{2}{3}.$$

- Marginally $\Pr(H_2) = 1/2$, so the flips are dependent
- Their joint distribution remains unchanged when we swap the flips
- Observing heads updates the shared unknown bias; one toss does not physically affect the next



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Pair discussion: learning about the coin

Discuss in pairs (1–2 minutes).

The coin's bias has prior $\theta \sim \text{Beta}(1, 1)$, and the flips are independent given θ .

Why does observing heads change our prediction for the next flip? Does this contradict conditional independence?



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The coin's bias has prior $\theta \sim \text{Beta}(1, 1)$, and the flips are independent given θ .

Why does observing heads change our prediction for the next flip? Does this contradict conditional independence?

Answer: We learn about the shared unknown bias. After heads, $\theta \mid H_1 \sim \text{Beta}(2, 1)$, changing the predictive probability from $1/2$ to $2/3$.

The coin has not changed. Independence holds when θ is fixed; averaging over an uncertain θ creates dependence.



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- Exchangeability does not mean that the results of the experiments could not be different
 - e.g. if we know that the experiments have been in two different laboratories, and we know that the other laboratory has better conditions for the rats, but we do not know which experiments have been made in which laboratory
 - a priori experiments are exchangeable
 - model could have unknown parameter for the laboratory with a conditional prior for rats assumed to come from the same place (clustering model)



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Exchangeability and additional information

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- Bioassay: doses x_j and group sizes n_j are fixed design information; y_j counts deaths



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Exchangeability and additional information

- Bioassay: doses x_j and group sizes n_j are fixed design information; y_j counts deaths
- Given α, β , the counts are independent, but generally not identically distributed:

$$y_j \mid \alpha, \beta, n_j, x_j \sim \text{Binomial}(n_j, \pi_j), \quad j = 1, \dots, J$$

where $\text{logit}(\pi_j) = \alpha + \beta x_j$.

$$p(\alpha, \beta \mid y, n, x) \propto p(\alpha, \beta) \prod_{j=1}^J p(y_j \mid \alpha, \beta, n_j, x_j)$$



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$$p(\alpha, \beta \mid y, n, x) \propto p(\alpha, \beta) \prod_{j=1}^J p(y_j \mid \alpha, \beta, n_j, x_j)$$

- Reordering complete records (x_j, n_j, y_j) leaves the likelihood unchanged; swapping only counts between doses generally changes it



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Hierarchical exchangeability

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- Rat example: experiments $j = 1, \dots, J$, each with n_j rats



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Hierarchical exchangeability

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- Rat example: experiments $j = 1, \dots, J$, each with n_j rats
- Within experiment j , outcomes are IID Bernoulli(θ_j) given θ_j ; they need not be exchangeable across experiments



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Hierarchical exchangeability

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- Rat example: experiments $j = 1, \dots, J$, each with n_j rats
- Within experiment j , outcomes are IID Bernoulli(θ_j) given θ_j ; they need not be exchangeable across experiments
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- Rat example: experiments $j = 1, \dots, J$, each with n_j rats
- Within experiment j , outcomes are IID Bernoulli(θ_j) given θ_j ; they need not be exchangeable across experiments
- The group probabilities θ_j are modelled exchangeably a priori
- Counts $y_j \mid \theta_j, n_j \sim \text{Binomial}(n_j, \theta_j)$ need not be exchangeable when n_j differs
- This assumes comparable experiments; grouping alone does not justify a shared population distribution



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- **Conditional exchangeability**

- Responses with the same covariates may be modelled exchangeably; different covariates may imply different distributions
- For fixed covariates, use a conditional model $p(y \mid x)$; adding x_i does not make (x_i, y_i) exchangeable



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- **Conditional exchangeability**

- Responses with the same covariates may be modelled exchangeably; different covariates may imply different distributions
- For fixed covariates, use a conditional model $p(y \mid x)$; adding x_i does not make (x_i, y_i) exchangeable

- **Partial exchangeability**

- Symmetry within specified groups, not under arbitrary exchanges across groups
- The models here additionally assume exchangeable group parameters, possibly given group-level covariates



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Section 3

Computational aspects



- Consider the **prior hierarchy**, before observing data:

$$\tau \sim p(\tau), \quad \tau > 0, \quad \theta_j \mid \tau \sim \mathcal{N}(0, \tau) \quad (\text{SD form})$$

- Small τ : group effects lie near 0; large τ : group effects are more spread out
- If plausible τ values range from near zero to large values, the joint prior has a narrow neck and a broad region: a funnel

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- Small τ : group effects lie near 0; large τ : group effects are more spread out
- If plausible τ values range from near zero to large values, the joint prior has a narrow neck and a broad region: a funnel
- With weak information about group effects, the **posterior can retain** this funnel geometry
- Not every hierarchical posterior is funnel-shaped: the likelihood also matters

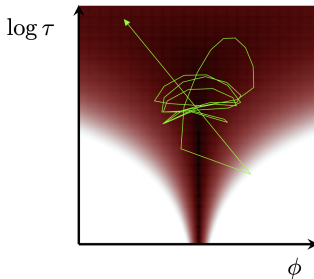
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Why is a funnel problematic?

- Changing curvature makes it hard for HMC to explore both the narrow and wide regions efficiently
- Post-warmup divergences warn of potentially unreliable exploration and biased posterior estimates
- Good $\hat{R}$ and ESS do not cancel divergence warnings



Betancourt (2020) [Stan diagnostics](#) [HMC demo](#)

In this figure, ϕ denotes one group effect θ_j .

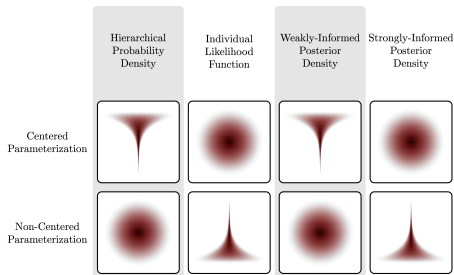


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Handling the funnel

- Raising `adapt_delta` targets smaller steps: it can help, but does not guarantee reliable inference
- **Centered**: often better with strong group information

$$\theta_j \sim N(\mu, \tau) \quad (\text{SD form})$$





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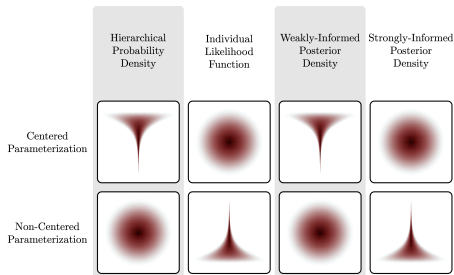
Handling the funnel

- Raising `adapt_delta` targets smaller steps: it can help, but does not guarantee reliable inference
- **Centered**: often better with strong group information

$$\theta_j \sim N(\mu, \tau)$$
 (SD form)

- **Non-centered**: often helps weakly informed group effects

$$\eta_j \sim N(0, 1), \quad \theta_j = \mu + \tau \eta_j$$





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Why non-centering preserves the model

Start with $\eta_j \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$, independent of the hyperparameters a priori, and set

$$\theta_j = \mu + \tau \eta_j, \quad \tau > 0.$$

Conditional on μ, τ , an affine transformation of a normal variable is normal, with

$$\begin{aligned} \mathbb{E}(\theta_j \mid \mu, \tau) &= \mu + \tau \mathbb{E}(\eta_j) = \mu, \\ \text{Var}(\theta_j \mid \mu, \tau) &= \tau^2 \text{Var}(\eta_j) = \tau^2. \end{aligned}$$

Thus the θ_j are conditionally independent normals with mean μ and SD τ , as in the centered hierarchy.

Keep the same hyperpriors and likelihood, evaluated at $\theta_j = \mu + \tau \eta_j$. Only the sampling coordinates change.



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Section 4

Additional material



Eight schools: historical SAT coaching study

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- Historical example from BDA3, Section 5.5
 - randomized coaching experiments in eight high schools
 - question: does short-term coaching improve verbal test scores?
 - estimate a coaching effect for each school



Eight schools: historical SAT coaching study

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- Historical example from BDA3, Section 5.5
 - randomized coaching experiments in eight high schools
 - question: does short-term coaching improve verbal test scores?
 - estimate a coaching effect for each school
- SAT-V in this study (the historical verbal test)
 - standardized multiple-choice test
 - score range 200–800, with mean about 500 and SD about 100
 - pre-test PSAT scores in verbal (V) and mathematics (M) help adjust for students' initial performance



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- Effectiveness of the SAT coaching
 - students took pre-tests PSAT-M and PSAT-V
 - some students received coaching
 - a regression in each school adjusts for pre-test scores and estimates the coaching effect y_j and its standard error σ_j
 - use a normal approximation for y_j , treating the estimated sampling variances σ_j^2 as known
 - y_j is an adjusted effect estimate, not a sample mean; σ_j is not the SD of individual students' scores



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- Data:

School	A	B	C	D	E	F	G	H
y_j	28	8	-3	7	-1	1	18	12
σ_j	15	10	16	11	9	11	10	18

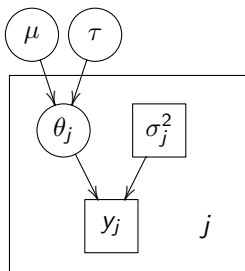


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Hierarchical normal model for school effects

$$\theta_j | \mu, \tau \sim \mathcal{N}(\mu, \tau)$$

$$y_j | \theta_j \sim \mathcal{N}(\theta_j, \sigma_j^2)$$



The observed y_j is a school effect estimate; σ_j is its reported standard error, treated as known.

Here $\mathcal{N}(\mu, \tau)$ is written in mean–SD form; τ is the between-school SD.



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Normal shrinkage: completing the square

For fixed μ and $\tau > 0$, with known sampling SD $\sigma_j > 0$,

$$y_j \mid \theta_j \sim \mathcal{N}(\theta_j, \sigma_j^2), \quad \theta_j \mid \mu, \tau \sim \mathcal{N}(\mu, \tau^2).$$

Here $\mathcal{N}$ uses variance as its second argument.

$$\begin{aligned} p(\theta_j \mid \mu, \tau, y) &\propto \exp \left[-\frac{1}{2} \left\{ \frac{(y_j - \theta_j)^2}{\sigma_j^2} + \frac{(\theta_j - \mu)^2}{\tau^2} \right\} \right] \\ &\propto \exp \left[-\frac{(\theta_j - m_j)^2}{2v_j} \right], \end{aligned}$$

where completing the square gives

$$v_j = \left(\frac{1}{\sigma_j^2} + \frac{1}{\tau^2} \right)^{-1}, \quad m_j = v_j \left(\frac{y_j}{\sigma_j^2} + \frac{\mu}{\tau^2} \right).$$

Thus $\theta_j \mid \mu, \tau, y \sim \mathcal{N}(m_j, v_j)$.

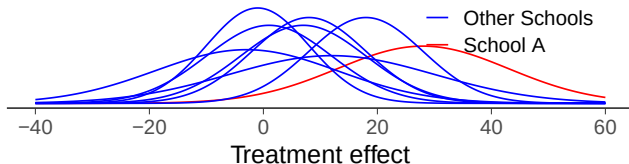


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Eight schools: pooling assumptions

How does the posterior for School A (red) change?

Separate model



No pooling: each school's effect uses its own data. School A's estimate is high but uncertain.

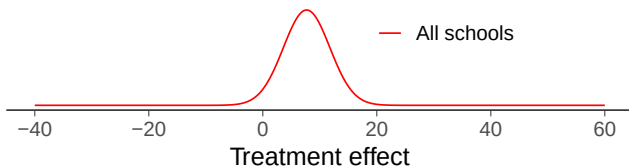
The hierarchical plots follow BDA3, Section 5.5, with flat independent priors on μ and $\tau > 0$, rather than the illustrative half-normal prior in the factory example.



Eight schools: pooling assumptions

How does the posterior for School A (red) change?

Pooled model



Complete pooling: all schools share a common effect. School A's posterior is identical to the others.

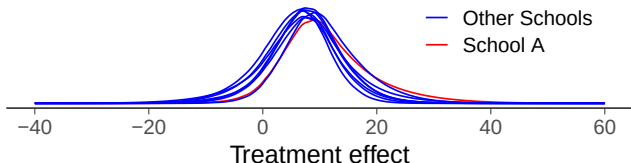
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Eight schools: pooling assumptions

How does the posterior for School A (red) change?

Hierarchical model



Partial pooling: the marginal posteriors $p(\theta_j | y)$ share information while allowing different effects.

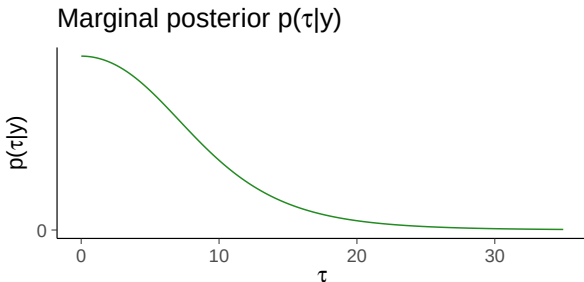
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Eight schools: uncertainty about variation

Does the peak near zero establish that all effects are equal?



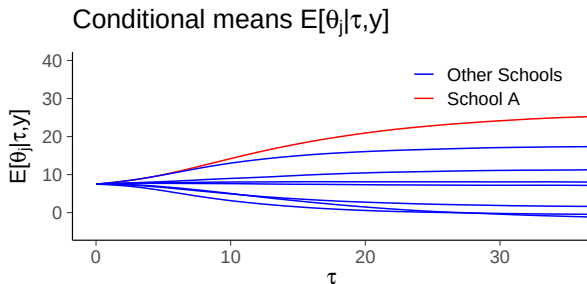
- This is $p(\tau | y)$, with μ and the school effects integrated out
- A range of between-school SDs remains plausible
- Full Bayes averages over this distribution; fixing τ at its mode would impose much stronger pooling



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Eight schools: conditional posterior means

What happens to the school estimates as τ changes?



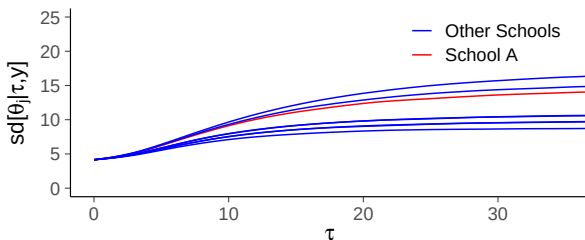
- Curves show $E(\theta_j | \tau, y)$: τ is fixed and μ is integrated out
- Small τ : means come together near the pooled estimate, 7.7
- Large τ : means approach the separate estimates y_j



Eight schools: uncertainty given τ

Does small τ make the common effect precisely known?

Conditional standard deviations $sd[\theta_j|\tau,y]$



- Curves show $sd(\theta_j | \tau, y)$, with uncertainty in μ included
- As $\tau \rightarrow 0$, effects coincide, but their common value is uncertain:

$$sd(\theta_j | \tau, y) \longrightarrow \left(\sum_{k=1}^J \frac{1}{\sigma_k^2} \right)^{-1/2} \approx 4.1.$$